YEAR 8
MATHS & SCIENCE

What pupils in England are expected to know, deepen and connect during Key Stage 3

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The crucial point

The government specifies the curriculum for Key Stage 3 (Years 7–9), not a single national Year 7 teaching order. Schools publish their own year-by-year curriculum and may sequence the statutory content differently.

Prepared as a clear reference for families supporting pupils through Year 8.

1. How to use this guide

This guide answers two different questions that are often confused:

- What does the Department for Education require pupils to learn during Key Stage 3?
- What should a parent or tutor watch for as pupils move through the middle of Key Stage 3?

The first is statutory. The second depends on the school's sequencing. For that reason, the curriculum maps below show the full KS3 entitlement, while the "Year 8 lens" identifies the kinds of consolidation, connection and increasing sophistication that support strong KS3 progression rather than claiming a government-mandated Year 8 order.

The government's three big mathematical aims

- Fluency — secure, flexible command of number, algebra and representations.
- Reasoning — making connections, testing conjectures, justifying and proving.
- Problem solving — choosing mathematics for routine and unfamiliar multi-step problems.

The government's three big science aims

- Scientific knowledge and conceptual understanding across biology, chemistry and physics.
- Understanding scientific methods, processes and enquiry.
- Using scientific knowledge to understand applications and implications of science.

Parent takeaway: success in Year 8 is not simply "covering more topics". It is about making previously learned ideas more secure, connecting them across topics, and using them with greater independence and precision.

2. Mathematics — the KS3 curriculum map

The statutory KS3 mathematics programme is organised into six content domains, underpinned by working mathematically. The concise descriptions below are faithful summaries of the DfE requirements.

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
NUMBER	Place value with integers and decimals; positive and negative numbers; fractions and mixed numbers; four operations; primes, factors, multiples, HCF/LCM and prime factorisation; powers and roots; standard form; fraction–decimal–percentage equivalence; percentages; units, rounding, estimation and calculator use.
ALGEBRA	Algebraic notation and vocabulary; substitution; simplifying and manipulating expressions; formulae and rearranging; linear equations; coordinates; graphs of linear and quadratic functions; gradients and intercepts; graphical solutions; sequences including arithmetic and geometric patterns and nth-term work.
RATIO & PROPORTION	Standard-unit conversions; scale factors, diagrams and maps; fractions as comparisons; ratio notation and division in a ratio; multiplicative relationships; links between ratio, fractions and linear functions; percentage change and simple interest; direct and inverse proportion; compound units such as speed, unit price and density.
GEOMETRY & MEASURES	Perimeter, area and volume; circles and composite shapes; scale drawings; ruler-and-compass constructions; geometric notation; congruence and similarity; transformations; angle facts and parallel lines; polygon angle sums; Pythagoras and introductory trigonometric ratios; 3-D properties and geometric reasoning.
PROBABILITY	Experimental frequency; fairness and equally/unequally likely outcomes; the 0–1 probability scale; probabilities summing to 1; systematic enumeration with tables, grids and Venn diagrams; sample spaces and theoretical probabilities for single and combined events.
STATISTICS	Representing and comparing distributions; discrete, continuous and grouped data; mean, median, mode, range and outliers; frequency tables and charts; bar and pie charts; interpreting relationships between two variables and representing them with scatter graphs.

Working mathematically — what sits underneath every topic

- Develop fluency: calculate accurately, use algebra confidently, and move between numerical, algebraic, graphical and diagrammatic forms.
- Reason mathematically: identify relationships, use ratio and proportion, test conjectures, begin deductive reasoning and judge what conclusions data can support.
- Solve problems: tackle multi-step and unfamiliar problems, use mathematics in financial contexts, model situations and select appropriate techniques.

3. Mathematics — the Year 8 progression lens

Because the DfE does not prescribe a national Year 8 order, use this as a progression and support map. By Year 8, pupils should increasingly move beyond isolated procedures and use number, algebra, ratio, geometry and data ideas together with greater fluency and reasoning.

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
NUMBER BECOMES A TOOL	Use integers, fractions, decimals, percentages, powers, roots, standard form, estimation and calculator methods flexibly inside multi-step problems rather than as isolated skills.
ALGEBRA BECOMES STRUCTURAL	Simplify and manipulate expressions, substitute reliably, solve equations, work with formulae, coordinates, sequences and graphs, and recognise the same relationship in symbolic, tabular and graphical forms.
PROPORTIONAL REASONING DEEPENS	Use ratio, scale, percentage change, rates and compound measures with increasing confidence; distinguish additive from multiplicative relationships and recognise direct/inverse proportional thinking.
GEOMETRY REQUIRES REASONING	Use angle facts, transformations, constructions, area/volume, similarity and developing deductive reasoning; explain why geometric conclusions follow, not merely quote an answer.
DATA & PROBABILITY BECOME EVIDENCE	Compare distributions using averages and spread, interpret graphs critically, construct sample spaces and reason about probabilities rather than treating charts as decoration.
MATHEMATICAL INDEPENDENCE	Select methods without being told which one to use, combine topics in unfamiliar problems, check reasonableness, communicate arguments clearly and correct errors intelligently.

A simple parent check

- Can your child explain a method, not merely produce an answer?
- Are fractions, decimals, percentages and negative numbers becoming dependable rather than fragile?
- Can they translate between words, tables, graphs, diagrams and algebra?
- Can they recognise when a problem is additive, multiplicative or proportional?
- Can they show enough working for someone else to follow their reasoning?

4. Science — the KS3 curriculum map

KS3 science is explicitly disciplinary: biology, chemistry and physics. Across all three, pupils must also work scientifically — planning, measuring, analysing, evaluating and communicating with increasing precision.

Working scientifically

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
SCIENTIFIC ATTITUDES	Objectivity; accuracy and precision; repeatability and reproducibility; how theories change with evidence; publication and peer review; risk evaluation.
EXPERIMENTAL SKILLS	Ask questions; make predictions; identify variables; select and carry out suitable enquiries; use apparatus safely; measure and record; evaluate methods; use sampling.
ANALYSIS & EVALUATION	Apply mathematics; present data in tables and graphs; identify patterns; draw conclusions; relate evidence to predictions; recognise random/systematic error; generate further questions.
MEASUREMENT	Use SI units and chemical nomenclature; use and derive simple equations; perform calculations; carry out basic statistical analysis.

5. Biology — what pupils must encounter across KS3

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
CELLS & ORGANISATION	Cells as the basic unit of life; microscopy; functions of major cell structures; plant versus animal cells; diffusion; unicellular adaptations; organisation from cells to tissues, organs, systems and organisms.
SKELETON & MUSCLES	Functions of the skeleton; biomechanics; forces produced by muscles; antagonistic muscle pairs.
NUTRITION & DIGESTION	Components of a healthy diet and their roles; energy needs; consequences of dietary imbalance; digestive organs and adaptations; enzymes as biological catalysts; gut bacteria; plant nutrition and water/mineral uptake.
GAS EXCHANGE	Human gas-exchange structures and functions; breathing using a pressure model; lung-volume measurements; effects of exercise, asthma and smoking; stomata in plants.
REPRODUCTION & HEALTH	Human reproductive systems, menstrual cycle at an introductory level, gametes, fertilisation, gestation and birth; maternal lifestyle and the foetus; plant reproduction, pollination, fertilisation, seed/fruit formation and dispersal; effects of recreational drugs.
PHOTOSYNTHESIS & RESPIRATION	Reactants/products and word equations; photosynthesis as the basis of food/energy stores and atmospheric balance; leaf adaptations; aerobic and anaerobic respiration, including fermentation, and their biological significance.
ECOLOGY, GENETICS & EVOLUTION	Food webs and interdependence; pollination and food security; environmental effects and toxic accumulation; heredity, chromosomes, genes and DNA; variation; natural selection; extinction; biodiversity and gene banks.

6. Chemistry — what pupils must encounter across KS3

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
PARTICLE MODEL	Solids, liquids and gases explained through particles; gas pressure; changes of state.
ATOMS, ELEMENTS & COMPOUNDS	A simple atomic model; distinguishing atoms, elements and compounds; symbols and formulae; conservation of mass in physical and chemical change.
PURE & IMPURE SUBSTANCES	Pure substances, mixtures and dissolving; diffusion; filtration, evaporation, distillation and chromatography; identifying purity.
CHEMICAL REACTIONS	Reactions as rearrangements of atoms; formulae and equations; combustion, decomposition, oxidation and displacement; acids and alkalis, pH and indicators; neutralisation; acid reactions with metals and alkalis; catalysts.
ENERGETICS	Qualitative energy changes during changes of state and in exothermic/endothermic reactions.
PERIODIC TABLE	Patterns in physical and chemical properties; Mendeleev's organising principles; groups and periods; metals and non-metals; using periodic patterns to predict reactions; metal/non-metal oxides and acidity.
MATERIALS	Reactivity series including carbon; extraction of metals from oxides using carbon; properties of ceramics, polymers and composites.
EARTH & ATMOSPHERE	Composition and structure of Earth; rock cycle; finite resources and recycling; atmosphere composition; human carbon dioxide production and climate impact.

7. Physics — what pupils must encounter across KS3

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
ENERGY	Energy in food and appliances; power ratings; energy transferred; domestic fuels, bills and resources; energy transfers by heating, radiation, motion, circuits, deformation, metabolism and combustion; conservation of energy in systems.
MOTION & FORCES	Speed = distance ÷ time; distance–time graphs; relative motion; contact/non-contact forces; force diagrams; balanced/unbalanced forces; moments; friction and drag; springs and Hooke's law; work and deformation; gravity, magnetism and static forces; pressure, upthrust and equilibrium.
WAVES & SOUND	Water waves, reflection and superposition; frequency in hertz; echoes, absorption and reflection; sound needing a medium; vibrations; longitudinal sound waves; hearing ranges; waves transferring energy and information.
LIGHT	Light in a vacuum; reflection, scattering and absorption; ray models for mirrors, pinhole cameras, refraction and convex lenses; the eye; light energy effects; colour, frequency, white light and prisms.
ELECTRICITY & MAGNETISM	Current in series and parallel circuits; charge flow; potential difference and resistance; conductors and insulators; static charge and electric fields; magnetic poles and field lines; Earth's field; electromagnets and basic DC motor principles.
MATTER	Physical versus chemical change; conservation and reversibility; density differences; Brownian motion; diffusion; particle arrangements/motion and changes of state; atoms and molecules; temperature, particle motion and internal energy.
SPACE	Weight and gravitational field strength; Earth–Moon–Sun gravity; Sun as a star, our galaxy and other galaxies; seasons and axial tilt; changing day length; light-year as astronomical distance.

8. Science — the Year 8 progression lens

As pupils move through Year 8, science should become increasingly explanatory and connected. Whatever order the school chooses, watch for pupils using models, evidence, mathematics and precise scientific language to explain unfamiliar situations rather than merely recalling facts.

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
FROM MODELS TO EXPLANATIONS	Use particle, cell, force, energy and wave models to explain observations, predict outcomes and recognise where a model has limits.
FROM VARIABLES TO BETTER ENQUIRY	Choose suitable enquiry approaches, control variables thoughtfully, improve procedures, consider sample size and distinguish repeatability from reproducibility.
FROM GRAPHS TO EVIDENCE	Use tables, graphs, equations and basic statistics to identify patterns, justify conclusions, recognise uncertainty and evaluate whether evidence really supports a claim.
FROM VOCABULARY TO PRECISION	Use scientific terminology, symbols, formulae, SI units and chemical nomenclature accurately enough that explanations become concise and testable.
FROM TOPICS TO BIG IDEAS	Connect structure and function in biology, particles and reactions in chemistry, and forces, energy, waves and fields in physics; notice where the same ideas reappear in different contexts.
FROM RECALL TO TRANSFER	Apply familiar scientific ideas to unfamiliar examples, explain rather than merely state, and use mathematics confidently inside scientific reasoning.

Questions parents can ask without turning home into another classroom

- What idea is this experiment testing?
- What did you change, measure and keep the same?
- What does the graph actually show?
- Which scientific model explains what happened?
- What evidence would make you change your explanation?
- Where did you use mathematics in this science topic?

9. The parent support plan — one page to remember

The aim is not to race ahead into GCSE content. It is to make Year 8 knowledge secure and connected enough that later KS3 learning can compound.

WATCH	SUPPORT	ASK
Maths fluency	Short, regular practice of any weak prerequisite that is blocking current Year 8 work	“Which earlier idea does this new topic depend on?”
Maths reasoning	Use mixed and unfamiliar problems that require pupils to choose the method	“Why does that method fit this problem?”
Science knowledge	Retrieve key vocabulary, models, equations and prior concepts across units	“What earlier science idea connects to this?”
Science enquiry	Discuss method quality, uncertainty, graphs and the strength of conclusions	“What makes this evidence convincing — or not?”
School alignment	Check the school’s published Year 8 curriculum each term	“Which KS3 strand are you doing now, and what prior knowledge does it build on?”

A useful rhythm: know what the school is teaching now, identify the prerequisite ideas, repair any gaps, then use retrieval, explanation and mixed practice to connect the new learning to what came before.

10. What to obtain from your child’s school

- The school’s Year 8 maths curriculum map or scheme overview.
- The school’s Year 8 science curriculum map or scheme overview.
- The order of units by term.
- How and when pupils are assessed.
- The school’s calculation, homework and revision expectations.
- Any set-specific or pathway-specific differences.

This matters because the DfE explicitly gives schools flexibility over when content is introduced within a key stage, while requiring schools to publish their curriculum year by year.

11. Official government sources used

No commercial curriculum, tutoring website, exam-board scheme or school scheme of work was used to define the curriculum content in this guide.

Department for Education. National curriculum in England: mathematics programme of study — key stage 3.

Ref: DFE-00179-2013

Published 11 September 2013; GOV.UK page updated 28 September 2021.

https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study

Department for Education. National curriculum in England: science programme of study — key stage 3.

Ref: DFE-00185-2013

Published 11 September 2013; GOV.UK page updated 6 May 2015.

https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study

Department for Education. National curriculum: secondary curriculum.

Ref: DFE-00183-2013

Published 11 September 2013. Statutory secondary programmes of study for Key Stages 3 and 4.

https://www.gov.uk/government/publications/national-curriculum-in-england-secondary-curriculum

Department for Education. Teaching mathematics at key stage 3.

Non-statutory guidance

Published 28 September 2021. Used only to clarify progression and curriculum planning, not to invent a Year 8 statutory syllabus.

https://www.gov.uk/government/publications/teaching-mathematics-at-key-stage-3

Final orientation

For a pupil in Year 8, the most useful mental model is this: Year 8 is the connecting middle movement of Key Stage 3, not a sealed syllabus of its own. The destination remains the full KS3 entitlement above; the school decides the route. Our job as parents and educators is to know the destination, inspect the school's route, strengthen weak prerequisites, and help the pupil connect ideas deeply enough for later KS3 and GCSE study.